University
Mohammed VI
Polytechnic

College of
Computing

Deliverable 6: DDL and DML

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	PRINT("MySQL")
Member 1	OUAHIDE Soufian
Member 2	SABER AbdelilLah
Member 3	OUSSAMA Yassine
Member 4	TIGAMI Imrane
Repository Link	`github.com/abdelillahsaber/DatabaseGroupCC`

---

1 Role of each team member

All four team members contributed to Part 1 (database definition) and Part 2 (population and modification operations). The remaining work was distributed as follows:

- **TIGAMI Imrane:** Worked with Soufiane on Sections A and B of Part 3 and was responsible for Part 4.
- **SABER Abdelilah:** Worked on Section D of Part 3, which combines joins and aggregation.
- **OUSSAMA Yassine:** Worked on Section C and two queries from Section E of Part 3.
- **OUAHIDE Soufiane:** Worked with Imrane on Sections A and B, and independently completed two queries from Section E and Section F of Part 3.

2 Main implementation difficulties encountered

The main difficulties encountered during this lab were:

1. **Maintaining consistency across the scripts.** Since different team members worked on different sections, some queries used inconsistent table or column names.
2. **Choosing appropriate nullability constraints.** We had to distinguish mandatory attributes from optional ones when deciding where to use `NOT NULL` and where to allow `NULL`.
3. **Understanding unfamiliar SQL constructs.** Some queries required constructs that we were less familiar with, such as `NOT EXISTS`. We consulted documentation and examples to understand and apply them correctly.

3 Assumptions

The following assumptions and methodological choices were made:

- **Unique email addresses.** We assumed that each person has a distinct email address. Therefore, we added a `UNIQUE` constraint to the `email` column of the `Person` table.
- **Default** status in `Reservation` is "Pending", `operational_status` in `EquipmentUnit` is "InService", `researchStatus` in `ResearchProject` is "Proposed".
- **ON DELETE CASCADE from Reservation to Reserves**, Justification: A `Reserves` record exists only because a reservation exists. If a reservation is deleted, its associated `Reserves` records should also be deleted because they no longer have any meaning without the reservation.

- **ON DELETE SET NULL for approverPid**, An approver is a person assigned to approve a reservation. If that person is deleted from the `Person` table, the reservation should not necessarily be deleted. Instead, `approverPid` should be set to `NULL`, indicating that the reservation no longer has a recorded approver.
- **Reproducible timestamp tests.** We used November 2026 as the reference month for the timestamp-filtering query and included corresponding dates in the dataset to make the tests reproducible.
- **AI-assisted database population.** As permitted by our teaching assistant, we used artificial intelligence to assist with generating the population data for Part 2.

4 Answers(Part4)

Requirement 1: Restricting attribute values:

We need to ensure that some attributes can only take specific values. For example, Program must be one of ('BSc', 'BEng', 'MSc', 'PhD'), and status_research in ResearchProject must be one of ('Proposed','Active','Suspended','Completed','Cancelled'). The SQL concepts studied so far do not allow us to enforce these restrictions because we have not yet studied CHECK constraints. The database needs to verify the value whenever a row is inserted or when the corresponding attribute is updated.

PR3 (Exactly one PI, at most one Co-PI, both Academic):

Each project must have exactly one PI and can have at most one Co-PI. Both the PI and Co-PI must be Academic members. The database needs to count the number of PIs and Co-PIs for each project and check whether they are Academic members whenever a participant is added, modified, or deleted. We cannot express all these conditions using the SQL concepts studied so far.

PR2/RE1 (Administrative staff cannot participate in ResearchProjects):

Any Person can participate in a ResearchProject or make a Reservation, except Administrative staff. We cannot enforce this rule using the SQL concepts studied so far because the database needs to check whether the person involved belongs to the Administrative table. This must be checked whenever a row is inserted or updated in Participates or Reservation.

CE7 (Only people holding the required certifications can reserve or use equipment units):

A Person can reserve or use an EquipmentUnit only if they hold all the certifications required by its EquipmentModel and those certifications are not expired. We cannot enforce this rule using the SQL concepts studied so far because the database needs to compare information from several tables (Reserves, EquipmentUnit, Requires, and Holds). This must be checked whenever a reservation is made and when it is approved.

CO10 (Consumables may be acquired only in the context of an existing Reservation):

A consumable must be acquired as part of an existing Reservation. We cannot enforce this rule using the SQL concepts studied so far because the database needs to check that the Reservation exists when the consumable is acquired.

PE6 (Each student is supervised by at least one academic staff member):

Every student must have at least one Academic advisor. We cannot enforce this rule using the SQL concepts studied so far because the database needs to check whether each student has at least one advisor in the Advises table. This must be checked when a student is added or when an advisor is removed.